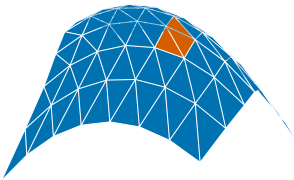
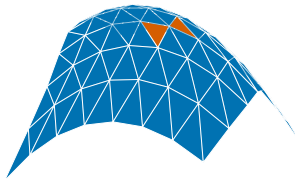
Frozen test-case gallery — synthetic examples

Background Inclusion

LOW ERROR · case 039
GT partition



Frozen PCA–Ridge prediction

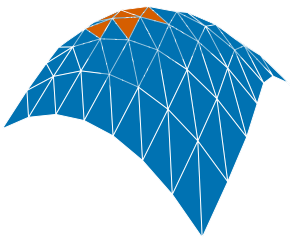


GT → prediction

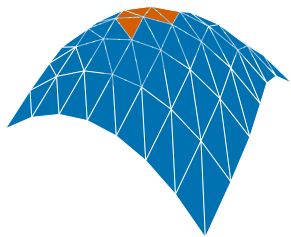
E_b: 6.32 → 6.33 kPa
ρ_E: 3.34 → 2.61

E_b relative error	**0.1%**
ratio relative error	**21.8%**
center normalized error	**9.0%**
radius relative error	**2.6%**

MID ERROR · case 004
GT partition



Frozen PCA–Ridge prediction

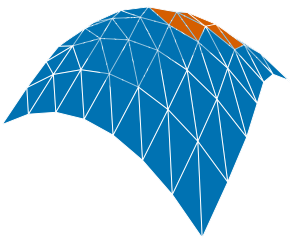


GT → prediction

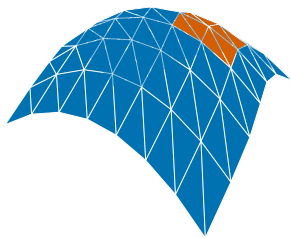
E_b: 2.81 → 2.69 kPa
ρ_E: 1.30 → 1.44

E_b relative error	**4.2%**
ratio relative error	**10.6%**
center normalized error	**5.2%**
radius relative error	**8.7%**

HIGH ERROR · case 134
GT partition



Frozen PCA–Ridge prediction



GT → prediction

E_b: 2.22 → 1.66 kPa
ρ_E: 2.60 → 5.67

E_b relative error	**25.0%**
ratio relative error	**118.3%**
center normalized error	**7.7%**
radius relative error	**2.0%**